

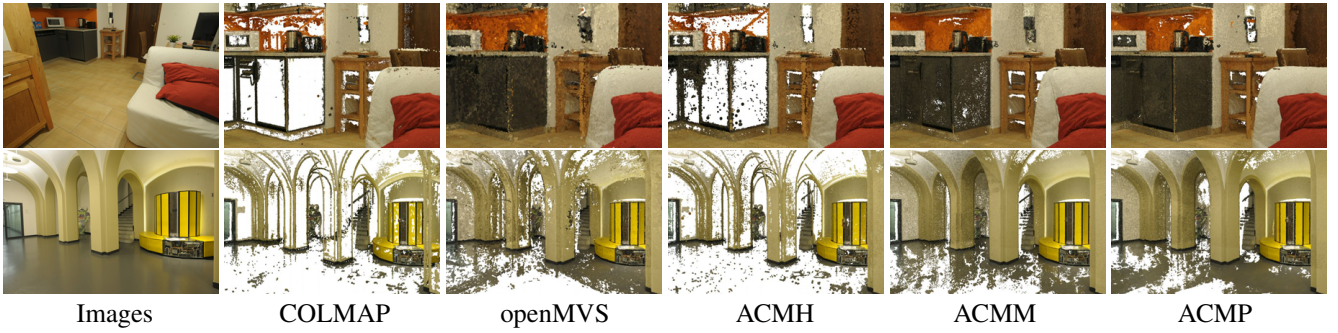


Figure 5: Qualitative point cloud comparisons of different MVS methods on living room and old computer of ETH3D benchmark. These 3D models are reported by the ETH3D benchmark evaluation server (?).

Table 2: Accuracy, completeness and F_1 score (in %) comparisons of reconstructed point clouds on the high-resolution multi-view test datasets of ETH3D benchmark at evaluation threshold $2cm$. The related values are from (?).

	method	Accuracy	Completeness	F_1
indoor	COLMAP	91.95	59.65	70.41
	openMVS	82.00	75.92	78.33
	ACMH	91.14	64.81	73.93
	ACMM	90.99	72.73	79.84
	ACMP	90.60	74.23	80.57
outdoor	COLMAP	92.04	72.98	80.81
	openMVS	81.93	86.41	84.09
	ACMH	83.96	80.03	81.77
	ACMM	89.63	79.17	83.58
	ACMP	90.35	79.62	84.36
all	COLMAP	91.97	62.98	73.01
	openMVS	81.98	78.54	79.77
	ACMH	89.34	68.62	75.89
	ACMM	90.65	74.34	80.78
	ACMP	90.54	75.58	81.51

record the running time of each stage in Table 3. As can be seen, the planar model construction occupies very little run-time. As for sparse correspondences generation, planar prior assistance and geometric consistency, their running time is very close. This is because these stages all employ the same pipeline of PatchMatch multi-view stereo. Therefore, with very little computational cost, our method without geometric consistency can achieve much better reconstruction results than other PatchMatch multi-view stereo methods that are implemented on the original image resolution.

Conclusion

In this work, we propose a planar prior assisted PatchMatch multi-view stereo framework to help the depth estimation in low-textured areas. To tackle depth ambiguities caused by unreliable photometric consistency, we leverage sparse credible correspondences to build planar models. These models can reflect the depth ranges of low-textured areas well but are still biased, especially for non-planar regions. Therefore, we embed these planar models into PatchMatch multi-view

Table 3: Running time of different stages of our method for an image of size 3200×2130 pixels on a single GPU.

Stage	Time (s)	Ratio (%)
Sparse Correspondences Generation	6.40	30.9
Planar Model Construction	1.08	5.2
Planar Prior Assistance	5.43	26.2
Geometric Consistency	7.79	37.6
Total Time	20.7	-

stereo by utilizing a probabilistic graphical model. This derives a novel multi-view aggregated matching cost, which jointly consists of photometric consistency and planar priors. This makes our method suited for both planar and non-planar regions. Experiments on ETH3D benchmark demonstrate the effectiveness of our methods by yielding state-of-the-art performance. We note that the performance of our method is comparable to ACMM. Since ACMM combines ACMH with the multi-scale geometric consistency, we think its performance gain mainly comes from the multi-scale geometric consistency framework. As a separate module, we have demonstrated that our method is much better than ACMH in our experiments. Thus, we may combine our method with the multi-scale geometric consistency to improve the reconstruction performance in the future work.

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